

Chi-Square Test: Concept, Examples, and Numerical Problems

1. Introduction

The Chi-Square (χ^2) Test is a non-parametric test used to analyze categorical data. It checks whether observed data matches expected data and helps in determining relationships between two categorical variables.

2. Types of Chi-Square Tests

Test Type	Purpose	Example
Chi-Square Goodness of Fit Test	Checks if the sample distribution fits an expected distribution	Do students prefer different types of soft drinks equally?
Chi-Square Test for Independence	Tests if two categorical variables are related	Is gender related to preference for online shopping?

3. Chi-Square Test Formula

The Chi-Square formula for both types of tests is:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

where:

- O = Observed frequency
 - E = Expected frequency
 - Σ = Summation of all categories
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4. Chi-Square Goodness of Fit Test Example

Problem 1: Checking if Student Preferences for Soft Drinks are Equal

A survey was conducted among **100 students** to check their preference for three soft drinks: **Coke, Pepsi, Sprite**. The observed frequencies are:

Soft Drink	Observed (O)
Coke	40
Pepsi	30
Sprite	30

If students had **no preference**, we expect each drink to be equally chosen ($E = 100/3 = 33.33$).

Step 1: Compute Chi-Square Value

$$\begin{aligned}\chi^2 &= \sum \frac{(O - E)^2}{E} \\ \chi^2 &= \frac{(40 - 33.33)^2}{33.33} + \frac{(30 - 33.33)^2}{33.33} + \frac{(30 - 33.33)^2}{33.33} \\ \chi^2 &= \frac{(6.67)^2}{33.33} + \frac{(-3.33)^2}{33.33} + \frac{(-3.33)^2}{33.33} \\ \chi^2 &= \frac{44.49}{33.33} + \frac{11.09}{33.33} + \frac{11.09}{33.33} \\ \chi^2 &= 1.33 + 0.33 + 0.33 = 2.0\end{aligned}$$

Step 2: Find Critical Value

- Degrees of freedom (df) = categories - 1 = 3 - 1 = 2.
- For $\alpha = 0.05$, critical $\chi^2 = 5.991$.

Step 3: Compare & Conclusion

Since $\chi^2 (2.0) < 5.991$, we fail to reject H_0 .

✅ **Conclusion:** Students have no strong preference among the three soft drinks.

5. Chi-Square Test for Independence Example

Problem 2: Is Gender Related to Shopping Preference?

A survey of **200 people** was conducted to check whether **gender (Male/Female)** is related to shopping preference (Online/In-Store). The observed data is:

	Online Shopping	In-Store Shopping	Total
Male	50	30	80
Female	70	50	120
Total	120	80	200

Step 1: Compute Expected Values

$$E = \frac{\text{Row Total} \times \text{Column Total}}{\text{Grand Total}}$$

	Online (E)	In-Store (E)
Male	$(80 \times 120)/200 = 48$	$(80 \times 80)/200 = 32$
Female	$(120 \times 120)/200 = 72$	$(120 \times 80)/200 = 48$

Step 2: Compute Chi-Square Value

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

$$\chi^2 = \frac{(50 - 48)^2}{48} + \frac{(30 - 32)^2}{32} + \frac{(70 - 72)^2}{72} + \frac{(50 - 48)^2}{48}$$

$$\chi^2 = \frac{4}{48} + \frac{4}{32} + \frac{4}{72} + \frac{4}{48}$$

$$\chi^2 = 0.083 + 0.125 + 0.056 + 0.083 = 0.347$$

Step 3: Compare with Critical Value

- $df = (\text{Rows} - 1) \times (\text{Columns} - 1) = (2 - 1) \times (2 - 1) = 1$.
- For $\alpha = 0.05$, critical $\chi^2 = 3.841$.

Step 4: Conclusion

Since $\chi^2 (0.347) < 3.841$, we fail to reject H_0 .

✅ Conclusion: No significant relationship exists between gender and shopping preference.

6. Additional Numerical Examples

Problem 3: Checking Fairness of a Dice Roll

A dice is rolled 120 times, and the outcomes are recorded:

Face	Observed (O)
1	18
2	22
3	15
4	20
5	25
6	20

- Expected frequency for each face (assuming fairness) = $120/6 = 20$.
- Apply the chi-square formula.

Problem 4: Testing Relationship Between Education and Job Preference

A survey of 150 people recorded job preference (Government vs. Private) based on education level (High School, Bachelor's, Master's). The table:

	Govt. Job	Private Job	Total
High School	40	30	70
Bachelor's	30	20	50
Master's	15	15	30

- Calculate expected values.
- Apply chi-square test.

Problem 5: Testing Preference for Smartphone Brands

A survey of 500 people recorded their brand preference:

Brand	Observed (O)
Apple	180
Samsung	140
OnePlus	100
Xiaomi	80

- If people had no preference, expected = $500/4 = 125$.
- Compute chi-square.

7. Conclusion

- **Chi-Square Goodness of Fit** → Tests if observed data matches expected data.
- **Chi-Square Test for Independence** → Tests if two categorical variables are related.
- If $\chi^2 > \chi^2_{critical}$, **reject H_0** (significant relationship).
- If $\chi^2 < \chi^2_{critical}$, **fail to reject H_0** (no relationship).